Initial Post - An Overview of Decision Trees and Random Forests

Decision Trees have a relatively simple structure that consists of nodes that branch off into a tree-like structure following a clearly defined pathway based on a sequence of decisions derived from the training data. Explainability is the key benefit of using these types of models which is emphasised by Bell (2020, Chapter 5) and Kneusel (2023, Chapter 3). This, of course, links back to the concept I mentioned in the first collaborative assignment when I stated that explainable AI is the key foundation for an AI model implementation.

However, the simplicity of the Decision Trees is a major limitation of this machine learning model. There is a risk that the data used will introduce bias into the model's output as is the case with Amazon's recruitment tool which introduced gender bias due to its use of historic data that was primarily from male candidates (Dastin, 2018). Furthermore, overfitting is an issue as the introduction of new data will result in the model performing poorly because it has learned the training data too precisely rather than generalising broader patterns.

Random Forests are essentially an extended Decision Tree; instead of one Decision Tree, we have a model that consists of many Decision Trees. Random Forests mitigate the risk of overfitting by introducing diversity into the model dataset (Theobald, 2024). The Random Forest combines the results of many Decision Trees in order to generate a more accurate result which is useful for credit decisioning and also may have mitigated the bias issues observed in the aforementioned Amazon recruitment tool.

Although more accurate, the main drawback of using the Random Forest method is that the model's explainability is reduced (Kneusel, 2023, Chapter 3; Theobald, 2024). This is due to the diversity—i.e. "randomness"—introduced by combining many Decisions Trees modelled off of different subsets of training data.

To conclude, while both Decision Trees and Random Forest models are commonly used machine learning algorithms, organisations need to weigh up the pros and cons of using either model to find the right balance between transparency and explainability against the need for improved predictive accuracy.

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Peer Response 1

Hi Emily,

Your post provides a comprehensive overview of Decision Trees (supervised learning) and K-Means Clustering (unsupervised learning).

Similar to my post, you examine the explainability of Decision Tree algorithms being the main reason they are used for fraud detection, credit scoring, and in the medical field for medical diagnosis. Thus, linking back to Leslie's (2019) key principles of Fairness, Accountability, Sustainability, and Transparency. Moreover, explainable AI should be central to any AI implementation (Peres *et al.*, 2020), which is epitomised by the use of Decision Trees.

The post addresses the drawbacks of using Decision Trees by mentioning that biased datasets can result in biased outcomes. For example, Amazon's biased recruitment tool, which was built on a data set that primarily consisted of data from male candidates (Dastin, 2018). I agree with the use of Random Forests to mitigate such bias in a model; however, a further exploration into the use of the Random Forest algorithm will enhance this post further; for instance, high training costs, show real-time predictions, and high memory storage (Reddy, 2025).

Your examination of the K-means clustering method was clearly outlined. Its application to customer segmentation is particularly interesting. This raises an additional question about its potential application in preferential pricing models within banking, where high-net-worth customers are often offered preferential interest rates or financial products.

Additionally, we could examine feature selection approaches to support the K-means clustering. Which method would be best for my example or really any use of K-means

clustering—manual or automatic? I suppose it really depends on the type of model being built and its use.

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Peer Response 2

Hi Sergei,

Thank you for your post that examines SVM and K-means clustering in the context of Predictive Maintenance (PM) in an industrial setting, with particular emphasis on the fact that algorithm selection is only one part of a wider end-to-end process.

I appreciate your discussion on SVM as a suitable algorithm for classifying machine states, while also indicating that these algorithms can generate results that are difficult to explain and interpret. SVMs are often considered black box models with limited transparency, as Bell (2020, Chapter 8) notes, "[i]n some cases you are left with a black-box implementation … but have little knowledge in between." Thus linking this to Leslie's (2019) AI implementation guiding principles, the lack of explainability emphasises the importance of transparency in AI models. Recent studies have attempted to overcome these challenges by using other machine learning algorithms, such as Decision Trees, to improve interpretability (Ramos Veira and Digiampietri, 2020). Including a discussion of these methods would further enhance this post.

Furthermore, this post examines the unsupervised k-means method and outlines that, although this model is fast to compute, its main drawback is the difficulty in selecting the k value. This is interesting in the sense that a poorly defined k value will result in a biased or random prediction. This begs the question: should a manual or automatic feature selection approach be used to support the k-means model?

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Peer Response 3

Hi Yasmin,

The use of credit card fraud in your post is a good example as it has been reported that financial losses from this type of fraud reached $33.41 billion in 2024 and are expected to continue rising over the coming years (Nilson, 2023, cited in Singh, Singh and Kumar, 2025).

I agree with your discussion on Decision Trees, particularly the aspects of transparency and the limitations associated with overfitting. A recent study by Flonder, Donath and Neamtu (2024) found that Decision Trees are a machine learning algorithm that is easy to implement, scale, and provide interpretable results; however, the paper also discusses the limitations of using synthetic data to train the model, which may differ from more complex real-world data. As you state, Random Forests can reduce the overfitting often found in Decision Trees. However, the data used remains important. If similar datasets are used across the Random Forest, this may not reduce bias and could instead reinforce it within the model. Echoing the work of Flonder, Donath and Neamtu (2024), we should consider sourcing more complex data that mirrors real-world data. Would you agree that sourcing more complex datasets to build a model would be difficult when using synthetic data?

Discussing the use of RNNs for fraud detection is also a good choice, as it complements the use of Decision Trees. Linking back to Isaac's point that recent studies have moved beyond using RNNs, several interesting experiments have emerged, such as using hybrid NN models (Jiang *et al.*, 2024) and CNN and LSTM networks (Chen, Y. *et al.*, 2025), which would be worth reviewing in order to contextualise your post further.

Overall, this post provides a clear overview of how Decision Trees and RNNs are used, while also acknowledging their limitations. Reviewing and incorporating more current research would further strengthen this discussion.

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Summary Post - An Overview of Decision Trees and Random Forests

Decision Trees and Random Forests are supervised machine learning algorithms that are trained to identify patterns in a dataset in order to generate a prediction. They are widely used in financial services (credit scoring/decisioning) and in the medical field to for diagnosing health conditions.

Decision Trees have a relatively simple structure that consists of nodes that branch off into a tree-like structure following a clearly defined pathway based on a sequence of decisions derived from the training data. Explainability is the key benefit of using these types of models which is emphasised by Bell (2020, Chapter 5) and Kneusel (2023, Chapter 3). This, of course, links back to the concept I mentioned in the first collaborative assignment when I stated that explainable AI is the key foundation for an AI model implementation.

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